From the Past

So far, we have been able to both solve some differential equations and check solutions to others.

- We solve $\frac{dy}{dt} = f(t)$, by _____ $f(t)$.
- We know that an nth order DE has ____ arbitrary constant in its general solution.
- We can use some known values of the solution to find the values of the arbitrary constants. This is called an _____.
- We know how to check if a given function $y(t)$ is a solution to a differential equation $\frac{dy}{dt} = f(t, y)$ by plugging $y(t)$ into both sides of the equation.

Question Solve the following initial value problems:

$\frac{dy}{dx} = 2x + 7$, $y(1) = 10$.

$\frac{d^2y}{dx^2} = \cos x - \sin x$, $y(\pi) = 2\pi$, $y'(\pi) = 1$.

Question What is the general solution of $\frac{dy}{dx} = 2y$? Find solutions with $y(0) = 1$, $y(0) = 0$, and $y(0) = -2$.

Today, we will look at a quick graphical way of examining multiple solutions to first-order DEs at once, without solving them.

Slope Fields

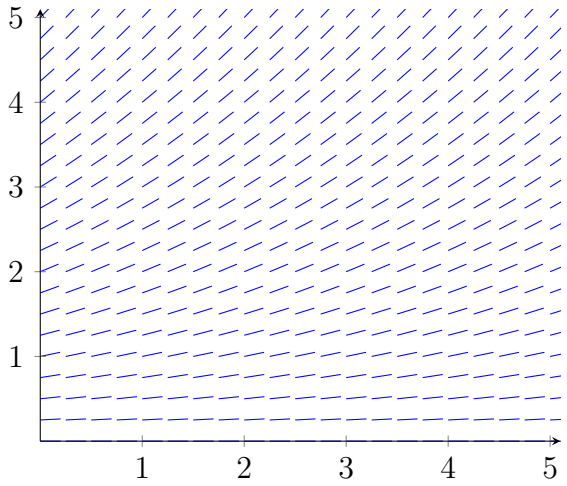
A general (first-order) differential equation can be written as

$$\frac{dy}{dt} = f(t, y),$$

where $f(t, y)$ is some function of t, y or both. This function gives the slope of the function $y(t)$ at the point (t, y).

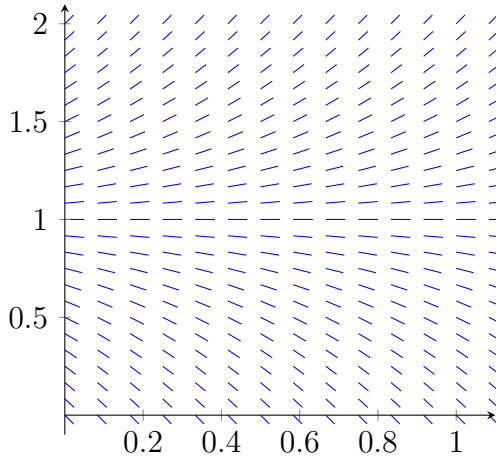
We can get an idea of the general behavior of a differential equation by looking at a plot of the slopes given by $f(t, y)$ at each point. These plots are called *slopefields*.
Consider the differential equation $\frac{dy}{dt} = 0.25y$. What is its general solution? (Hint: see similar questions on the previous worksheet.)

Slopefields allow us to look at the solution a different way. Below is the slopefield for $\frac{dy}{dx} = 0.25y$, a plot of the slopes, which are represented by short line segments. Sketch the solutions corresponding to $y(0) = 0.5$, 1, and 3.



Notice that if you move horizontally in the slopefield above, all the ticks have the same slope, whereas if you move up and down, the slope changes. Why is this?

1. Let's look now at $\frac{dy}{dt} = 2(y-1)$ and the initial conditions $y(0) = 0.75$, and $y(0) = 1.25$ □Regrade on the slopefield below.



(a) Draw solutions for each of those initial conditions on the slope field.

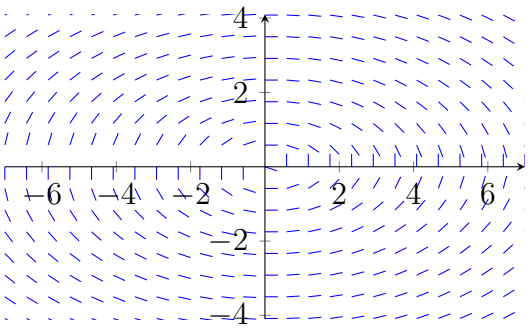
(b) What happens to each one of these solutions at $t \to \infty$? Answer from your drawings!

(c) What if we have the initial condition $y(0) = 1$? Draw a solution corresponding to such an initial condition. Can you write down a formula for it?

(d) Note that the slopes in the slopefield only change when we move _____, but not when we move _____. Can you explain this?

2. Below is a slopefield for the differential equation $\frac{dy}{dt} = -\frac{t}{3y}$.

☐Regrade



(a) In the first quadrant the slopes are positive/negative (cross out the wrong answer).

(b) What about in the other quadrants? Check that this matches up with the differential equation!

(c) Sketch solutions for each of the following initial conditions: $y(0) = 2$, $y(0) = 4$, $y(3) = 0$, $y(-4) = 1$. Be sure to label which is which.

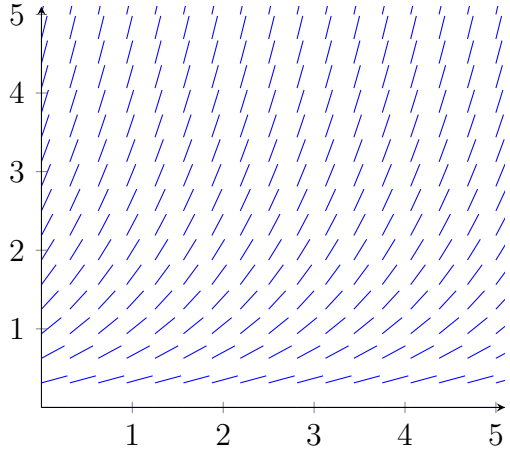
(d) What geometric figure do the solution curves look like?

(e) Are there any constant solutions, such as $y = 1$ was in the previous question? How can you tell just from the differential equation itself?

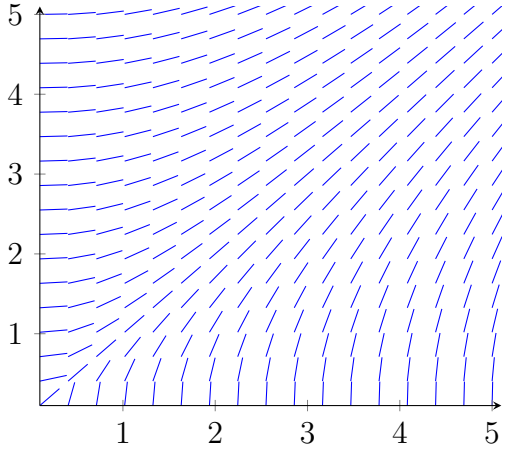
(f) Notice that in this case, the slopes are not constant if you move horizontally *or* vertically. Why not? How could you tell this from the differential equation itself?

3. For the slope fields below identify the differential equation and sketch three solutions for three different initial conditions. □Regrade

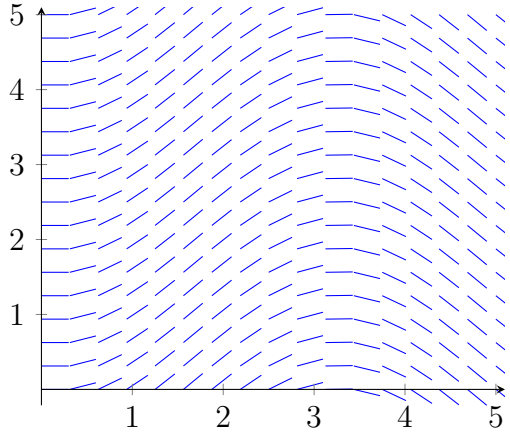
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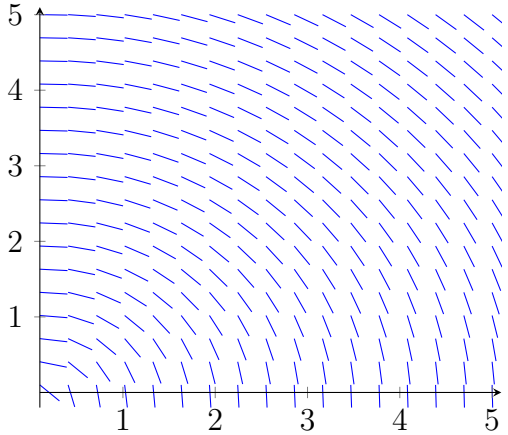
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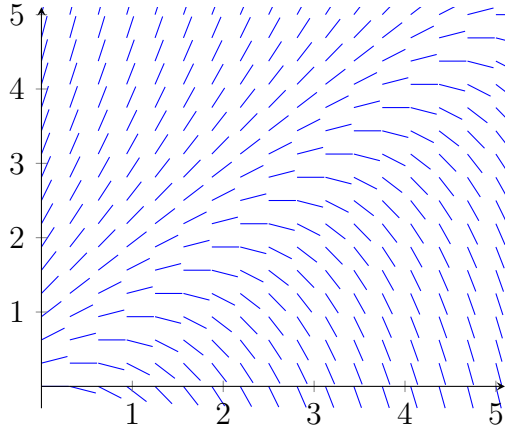
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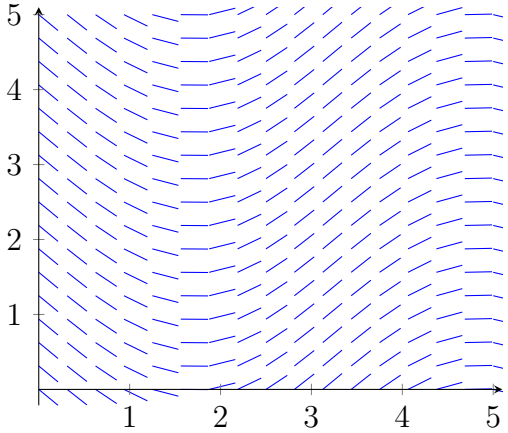
v)



iii)



vi)



(a) $\frac{dy}{dt} = \frac{t}{y}$ _____ (b) $\frac{dy}{dt} = y$ _____ (c) $\frac{dy}{dt} = \sin(t)$ _____

(d) $\frac{dy}{dt} = -\frac{t}{y}$ _____ (e) $\frac{dy}{dt} = y - t$ _____ (f) $\frac{dy}{dt} = -\cos(t)$ _____

Important Principle Notice that in all the above slopefields, solutions to the differential equations with different initial values *do not cross each other*. This will be true of all the differential equations we study. Keep this in mind when you do any work with solutions of differential equations!